

8. (a) For a number of years CFCs were used in refrigeration and in air conditioning. The use of these compounds is in decline as they damage the ozone layer. A typical CFC is chlorotrifluoromethane.

Give an equation that shows the formation of a chlorine radical from chlorotrifluoromethane. [1]

- (b) CFCs are being replaced by hydrofluorocarbons (HFCs) such as difluoromethane, which can be made by the reaction of dichloromethane with hydrogen fluoride in the presence of a catalyst.



One problem with this reaction is that chlorofluoromethane is also produced.

The table below shows the percentage of the organic products formed (and unreacted dichloromethane) for various starting ratios of the reactants.

Mole ratio HF : CH ₂ Cl ₂	Percentage composition of mixture / %		
	CH ₂ Cl ₂	CH ₂ ClF	CH ₂ F ₂
26 : 1	1.5	6.5	92.0
22 : 1	1.8	10.2	88.0
19 : 1	3.0	11.0	86.0
16 : 1	4.5	11.5	84.0



- (i) Write a statement that links the mole ratio of $\text{HF} : \text{CH}_2\text{Cl}_2$ with the yield of difluoromethane. [1]

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- (ii) In an experiment using 0.040 mol of CH_2Cl_2 , the yield of CH_2F_2 was 89.0%. Calculate the mass of difluoromethane produced. [2]

Mass = g

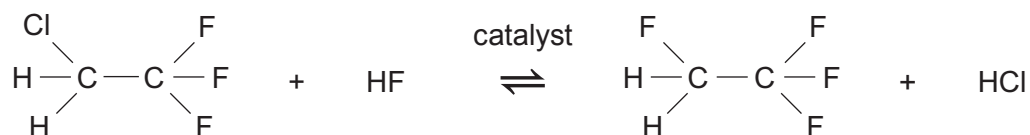


- (c) 1,1,1,2-Tetrafluoroethane is an HFC that is made from trichloroethene and hydrogen fluoride in a two-stage vapour phase process.

Stage 1 – an exothermic process



Stage 2 – an endothermic process



- (i) Suggest, giving a reason, whether a high or low pressure should be used in **stage 1**. [1]

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- (ii) Suggest, giving a reason, whether a high or low temperature should be used in **stage 2**. [1]

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- (iii) The boiling temperatures of the organic products formed after each stage are shown below.

Compound	Boiling temperature / °C
1,1,1-trifluoro-2-chloroethane	7
1,1,1,2-tetrafluoroethane	-26

The final product tends to be contaminated with some of the chlorocompound formed in the first stage.

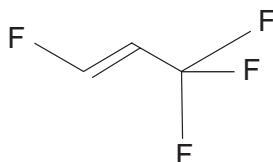
State the name of the method that can be used to separate these two compounds and suggest how this could be carried out in practice. [2]

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- (d) The use of HFCs is now declining as these compounds are contributing to global warming. Compounds such as HFOs (hydrofluoroolefins) are under development as suitable replacements. One such HFO is 1,3,3,3-tetrafluoropropene.



- (i) Explain why 1,3,3,3-tetrafluoropropene exists as *E/Z* isomers. [1]

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- (ii) The addition of hydrogen bromide to 1,3,3,3-tetrafluoropropene results in two structural isomers, each of molecular formula $C_3H_3BrF_4$.

I. State the type of mechanism occurring in this reaction. [1]

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II. These two compounds are formed in **approximately equal amounts** in this reaction.

Suggest why this ratio is different to that found in the reactions of other alkenes. [1]

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- (iii) Both the compounds formed in part (ii) will rotate the plane of plane polarised light.

Give the formula of **both** compounds indicating the chiral centre in both. [2]

- (iv) Draw the two mirror image forms of the compound 1-bromo-1,2-difluoroethane, $CHBrFCH_2F$. [1]

